Fundamentals of Data Science

Study Notes & Question Bank

Unit I — Introduction to Data Science
Unit II — Data Collection and Pre-processing

Unit I — Introduction to Data Science

Q1. Define Data Science. Explain its scope and importance in various fields.

Answer:

Definition:

Data Science is the study of data to extract useful information and knowledge. It uses techniques from statistics, mathematics, programming and domain knowledge to analyze data and solve real world problems.

Scope of Data Science:

- It includes data collection, cleaning, analysis, visualization and interpretation.
- It uses tools like Python, R, SQL, Machine Learning, and Data Visualization.
- It helps in predicting future trends and making better decisions.

Importance in Various Fields:

1. Healthcare: Helps in disease prediction, patient care and medical research.
2. Finance: Used in fraud detection, risk management, and investment decisions.
3. Retail: Helps in customer behavior analysis, sales prediction and product recommendation.
4. Education: Helps in student performance analysis and improving learning methods.
5. Manufacturing: Used in quality control, production planning and maintenance.
6. Government: Helps in policymaking, data driven decisions and public services improvement.

> ***Conclusion:*** *Data Science is a powerful tool that helps organizations and industries to make smart decisions, improve efficiency and solve complex problems.*

Q2. Explain the history and evolution of Data Science.

Answer:

Data Science is not a new field. It has developed over many years by combining statistics, computer science and domain knowledge.

History and Evolution:

1. 1960s — Early Beginnings:
    - The foundation of Data Science started with statistics, mathematics and early computers.

- Data was collected and analyzed using simple statistical methods.

2. **1970s** — Data Management:
   - Large organizations started using databases to store and manage data.
   - Structured Query Language (SQL) was developed to work with data.
3. **1980s** — Growth of Computing:
   - Personal computers made data processing easier.
   - Spreadsheet tools like Excel were widely used for data analysis.
4. 1990s — Data Mining Emerges:
   - Term "Data Mining" was introduced.
   - Organizations started finding useful patterns from large datasets.
5. **2000s** — Big Data Era:
   - Huge amounts of data started coming from internet, mobile phones and sensors.
   - Technologies like Hadoop and distributed systems were developed to handle big data.
6. **2010s** — Rise of Modern Data Science:
   - Machine Learning, Artificial Intelligence and Cloud computing became popular.
   - Python and R languages became main tools for data analysis.
7. 2020s — Present and Beyond:
   - Data Science is now used in almost every field.
   - Focus is on real-time data, automation and ethical use of data.

> ***Conclusion:*** *Data Science has evolved from simple statistical analysis to a powerful field that helps in decision making, prediction and solving real world problems.*

Q3. Explain the major applications of Data Science.

Answer:

Data Science is used in many fields to solve problems, make better decisions and improve efficiency. Some of the major applications are explained below.

1. Healthcare

- Data Science helps in predicting diseases, patient care and treatment.
- It is used to analyze medical images, test reports and health records.
- It also helps in drug discovery and monitoring of patients.

2. Finance and Banking

- It is used for fraud detection and risk management.
- Helps in credit scoring, loan prediction and customer segmentation.
- Stock market analysis and financial forecasting are also done using Data Science.

3. Retail and E-commerce

- Used to understand customer behaviour and buying patterns.
- Helps in product recommendation and personalized offers.
- Inventory management and demand forecasting are also done.

4. Education

- Helps in analyzing student performance and learning patterns.
- Used for identifying weak areas and suggesting improvements.
- Helps in predicting student placements and dropouts.

5. Manufacturing and Industry

- Used for quality control and defect detection.
- Helps in predicting machine failures and maintenance.
- Improves production efficiency and reduces costs.

6. Transportation and Logistics

- Used for route optimization and reducing travel time.
- Helps in demand forecasting and fleet management.
- Improves fuel efficiency and customer satisfaction.

7. Social Media and Marketing

- Helps in analyzing user behaviour and trends.
- Used for targeted advertising and sentiment analysis.
- Helps companies to improve their products and services.

8. Agriculture

- Used to predict crop yield and soil health.
- Helps in weather forecasting and pest detection.
- Improves farming productivity and reduces losses.

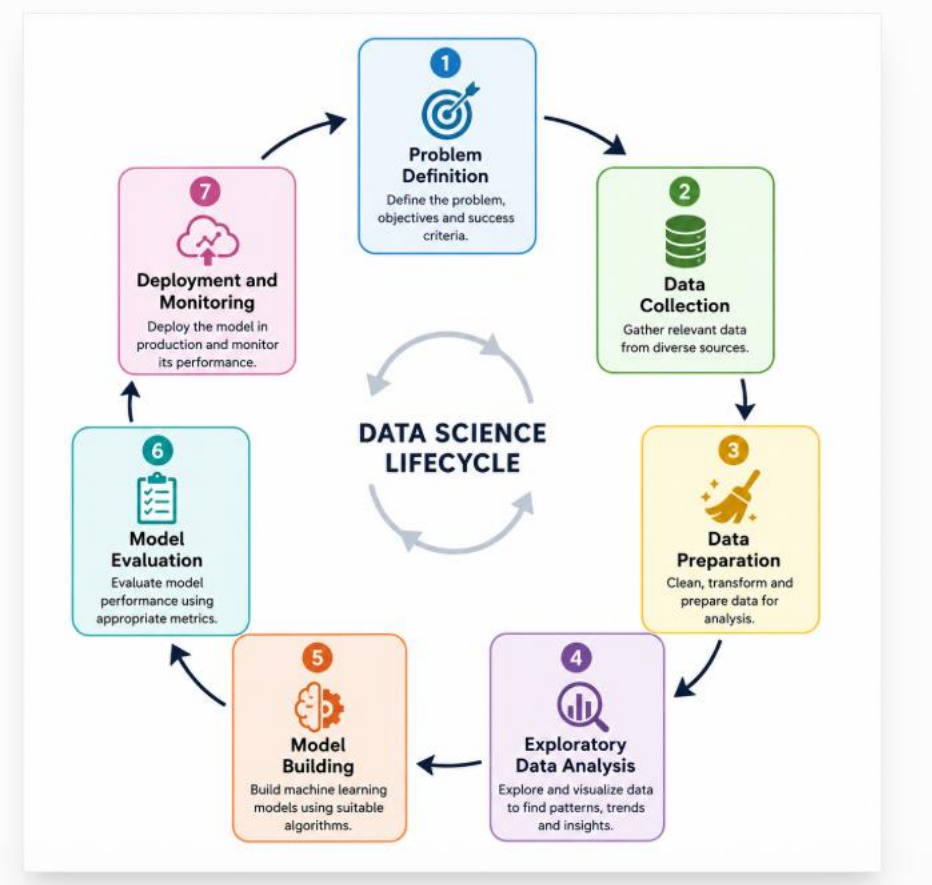
> ***Conclusion:*** *Data Science is useful in almost every field. It helps in analyzing large amounts of data and provides valuable information for better decisions, efficiency and growth.*

Q4. Explain the Data Science Lifecycle with a neat diagram and describe each stage.

Answer:

The Data Science Lifecycle is a step-by-step process used to solve problems and get useful insights from data. It is a cyclic process consisting of seven stages:

1. Problem Definition → 2. Data Collection → 3. Data Preparation → 4. Exploratory Data Analysis → 5. Model Building → 6. Model Evaluation → 7. Deployment and Monitoring → (back to Problem Definition, forming a repeating cycle)



Description of Each Stage:

1. Problem Definition: In this stage, we understand the problem, define the objective and decide what kind of data and solution is needed.
2. Data Collection: In this stage, we collect data from various sources like databases, files, sensors, websites, APIs, etc.

3. Data Preparation: In this stage, we clean the data by removing missing values, duplicates and handling errors. We also transform data into the right format for analysis.
4. Exploratory Data Analysis (EDA): In this stage, we explore and analyze the data using charts, graphs and summary statistics to find patterns, trends and relationships.
5. **Model Building**: In this stage, we choose the right algorithms and build models using the prepared data.
6. **Model Evaluation:** In this stage, we test the model using evaluation metrics to check how well the model performs.
7. Deployment and Monitoring: In this final stage, we deploy the model into real-world applications and monitor its performance. We update the model if needed.

> ***Conclusion:*** *The Data Science Lifecycle helps us solve problems in a systematic way and convert data into valuable insights for better decision making.*

Q5. Compare arrays, lists, matrices, and Data Frames with suitable examples.

Answer:

Arrays, lists, matrices and Data Frames are important data structures used in Data Science. They differ in their structure, type of data they store and use.

DATA STRUCTURES IN PYTHON – COMPARISON

Feature	Arrays (NumPy)	Lists	Matrices (NumPy 2-D Array)	Data Frames (Pandas)
Definition	A collection of elements of the same data type stored in contiguous memory.	An ordered collection of elements which can be of different data types.	A 2-D array with rows and columns where all elements are of the same data type.	A 2-D labeled data structure with columns of possibly different data types.
2. Dimensions	1-D	1-D (or nested)	2-D	2-D
3. Data Type	Homogeneous (same type)	Heterogeneous (different types allowed)	Homogeneous (same type)	Heterogeneous (different types in columns)
4. Indexing	Accessed using integer index.	Accessed using integer index (can be nested).	Accessed using row and column index (i, j).	Accessed using row index and column name.
5. Mutability	Usually fixed size (depends on language).	Size is dynamic (can grow or shrink).	Fixed size (after creation).	Size is dynamic (rows and columns can be added).
6. Use	Efficient for storing and processing large numeric data.	Useful when we have mixed data or need flexible structure.	Used for mathematical operations like addition, multiplication.	Used for real world tabular data analysis and manipulation.

7. Example (in Python)	`import numpy as np`
	`arr = np.array([1,2,3,4])`
	`# Output:`
	`[1 2 3 4]`
		`lst = [10, 'apple', 3.5, True]`
		`# Output:`
		`[10, 'apple', 3.5, True]`
			`import numpy as np`
			`mat = np.array([[1, 2, 3],`
			`[4, 5, 6]])`
			`# Output:`
			`[[1 2 3]`
			`[4 5 6]]`
				`import pandas as pd`
				`data = {`
				`'Name' : ['Ram', 'Sai'],`
				`'Age' : [20, 21],`
				`'Marks': [85, 90]`
				`}`
				`df = pd.DataFrame(data)`
				`# Output:`
				`Name Age Marks`
				`0 Ram 20 85`
				`1 Sai 21 90`

1-D Array

1	2	3	4

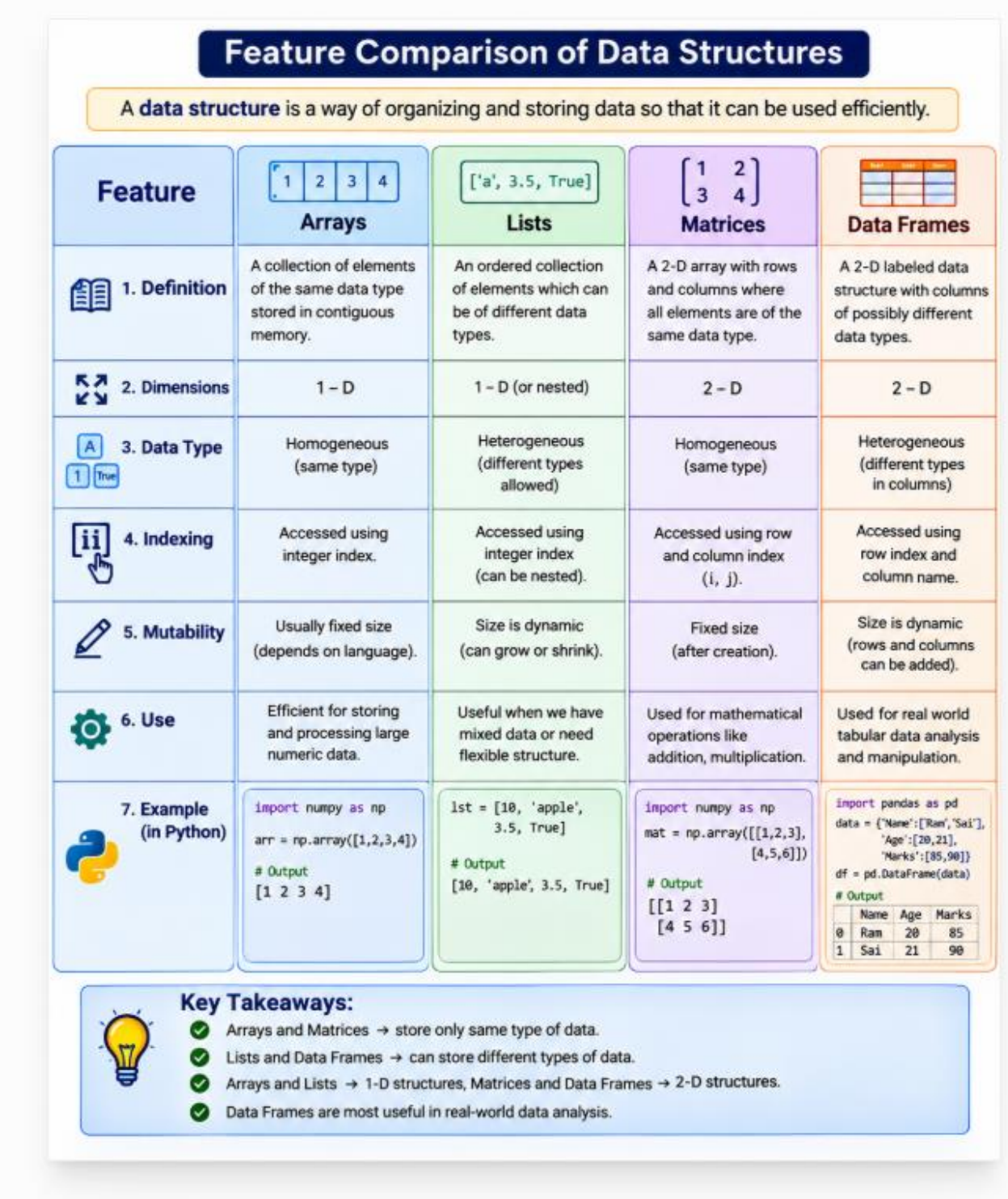
List

10	'apple'	3.5	True

2-D Matrix

1	2	3
4	5	6

	Name	Age	Marks
0	Ram	20	85
1	Sai	21	90



Conclusion:

- Arrays are best for storing large numeric data of same type.
- Lists are flexible and can store different types of data.
- Matrices are mainly used for numerical and mathematical operations.

- Data Frames are powerful for handling real world tabular data with rows and columns.

Q6. Explain the important tools and technologies used in Data Science.

Answer:

Data Science uses many tools and technologies to collect, process, analyze and visualize data. The important ones are explained below.

1. Programming Languages

- Python — Most popular for data analysis, machine learning, visualization (Libraries: NumPy, Pandas, Scikit-learn, Matplotlib).
- R — Used for statistical analysis and data visualization.

2. Data Collection Tools

- Web Scraping (BeautifulSoup, Scrapy) — To collect data from websites.
- APIs — To collect data from online services.

3. Data Storage Technologies

- SQL — For storing and managing structured data in relational databases (MySQL, PostgreSQL).
- NoSQL — For unstructured or semi-structured data (MongoDB, Cassandra).

4. Data Processing and Analysis Tools

- Pandas (Python) / dplyr (R) — For data cleaning, manipulation and analysis.
- Excel — Used for basic data processing and analysis.

5. Machine Learning Libraries

- Scikit-learn (Python) — For building machine learning models.
- TensorFlow / Keras — For deep learning applications.
- PyTorch — Another deep learning framework.

6. Data Visualization Tools

- Matplotlib, Seaborn (Python) — For static and advanced visualizations.
- Tableau, Power BI — For interactive dashboards and business intelligence.

7. Big Data Technologies

- Hadoop — For storing and processing large datasets.
- Apache Spark — For fast data processing and real-time analytics.

8. Cloud Platforms

- AWS, Google Cloud, Microsoft Azure — Provide storage, computing power and machine learning services in the cloud.

9. Other Useful Tools

- Jupyter Notebook — For writing and sharing code, results and visualizations.
- Git & GitHub — For version control and collaboration.

Conclusion: *These tools and technologies help data scientists to work efficiently with data and build useful solutions for real-world problems.*

Q7. Write a Python program to import a dataset using Pandas and perform basic exploration using head(), tail(), shape, info(), and describe().

Answer:

```
# Python program to import a dataset and perform basic exploration
import pandas as pd

# 1. Import dataset
df = pd.read_csv('sample_dataset.csv')   # load CSV file

# 2. Basic exploration
print("First 5 rows of the dataset (head):")
print(df.head())                              # shows first 5 rows

print("\nLast 5 rows of the dataset (tail):")
print(df.tail())                               # shows last 5 rows

print("\nShape of the dataset (rows, columns):")
print(df.shape)                               # returns (number of rows, number of columns)

print("\nInformation about the dataset:")
df.info()                                        # gives info about columns, data types, non-null values

print("\nStatistical summary of numeric columns:")
```

```
print(df.describe())                          # gives count,
mean, std, min, 25%, 50%, 75%, max
```

Explanation:

- pandas is imported as pd.
- The dataset is loaded using read_csv() function.
- head() displays the first 5 rows of the dataset.
- tail() displays the last 5 rows.
- shape returns the number of rows and columns.
- info() gives details like column names, data types, and non-null counts.
- describe() gives statistical summary (count, mean, std, min, quartiles, max) for numeric columns.

Unit II — Data Collection and Pre-processing

Q8. Explain different methods of data collection with suitable examples.

Answer:

Data collection is the process of gathering information from various sources. The different methods of data collection are:

1. **Observation Method:** In this method, data is collected by observing people, events or objects.
   - Example: A researcher observes customers in a shopping mall to know their buying behavior.
2. **Interview Method**: In this method, data is collected by asking questions to individuals.
   - Example: A HR manager interviews job candidates to assess their skills.
3. **Questionnaire Method:** In this method, data is collected using a set of written questions.
   - Example: A survey form given to students to know their study habits.
4. **Survey Method:** In this method, data is collected from a large number of people by visiting them or using online forms.
   - Example: A company conducts a market survey to know customer satisfaction.
5. **Secondary Data Collection:** In this method, data is collected from already available sources like books, websites, reports, journals, etc.
   - Example: Using census data from government reports.

> ***Conclusion:*** *These methods help to collect accurate and useful data for analysis and decision making.*

Q9. What is data pre-processing? Explain the major steps involved in data pre-processing.

Answer:

Data pre-processing is the process of cleaning and preparing raw data to make it suitable for analysis and modeling. Raw data may contain errors, missing values, duplicate data, or be in an inconsistent format. Pre-processing improves the quality of data and helps in getting accurate results.

Major Steps in Data Pre-processing:

1. **Data Cleaning:** In this step, we remove or correct noisy, missing, inconsistent and duplicate data.
    - Example: Removing duplicate records in a student list.
2. **Handling Missing Values:** In this step, missing or undefined values are handled. They can be filled using methods like mean, median, mode or by removing the record.
    - Example: Filling missing age values with the average age.
3. **Data Transformation:** In this step, data is converted into a proper format or scaled to make it suitable for analysis.
    - Example: Converting text data into numbers or normalizing values between 0 and 1.
4. **Data Integration:** In this step, data from multiple sources is combined into a single dataset.
    - Example: Combining sales data from online and offline stores.
5. **Data Reduction:** In this step, the size of data is reduced while keeping important information.
    - Example: Removing unnecessary attributes or using sampling.

> ***Conclusion:*** *These steps help in improving data quality and make it ready for analysis and machine learning.*

Q10. Explain data cleaning techniques with examples.

Answer:

Data cleaning is the process of finding and correcting or removing errors, inconsistencies and irrelevant data from a dataset. The main data cleaning techniques are:

1. **Handling Missing Values:** Missing values occur when some data is not available. We can fill them using mean, median, mode or by removing that record.
    - Example: In a student marks table, if maths marks is missing, we can fill it using average of that class.
2. **Removing Duplicates:** Duplicate records are removed to avoid repetition.
    - Example: If a customer record appears two times, we keep only one record.
3. **Correcting Inconsistent Data:** Sometimes data is written in different formats or contains spelling mistakes. We correct it to make it consistent.

- Example: “Hydrabad”, “HYDERABAD”, “Hyderabad” are changed to “Hyderabad”.

4. **Removing Irrelevant Data:** Irrelevant or unnecessary data that is not useful for analysis is removed.
   - Example: Removing “middle name” column if it is not needed for analysis.
5. **Detecting and Removing Outliers:** Outliers are values that are very high or very low compared to other values. They can affect results.
   - Example: In age column, if most ages are between 18-30 and a value is 150, it is an outlier and can be removed.

Conclusion: *Data cleaning improves the quality of data and helps in accurate analysis and better decision making.*

Q11. Explain different techniques for handling missing values with an example.

Answer:

Missing values occur when some data is not available. These missing values can affect the accuracy of analysis. So, we handle them using different techniques. The main techniques are:

1. **Deletion (Listwise deletion):** Remove the entire record (row) if it contains a missing value. This is simple but may result in loss of important data.
   - Example: If age is missing for a student, we can remove that student record.
2. **Deletion (Pairwise deletion):** Use available data for analysis and ignore the missing value for that specific calculation.
   - Example: While calculating average age, ignore records where age is missing.
3. **Mean / Median Imputation:** Replace missing numeric values with the mean or median of that column.
   - Example: If age is missing, fill it with average age of all students.
4. **Mode Imputation:** Replace missing categorical values with the mode (most frequent value).
   - Example: If city is missing, fill it with the most common city.
5. **Interpolation:** Estimate missing numeric values using nearby values.
   - Example: If one month's sales figure is missing, estimate it using sales of previous and next months.

6. Advanced Methods: Use machine learning or statistical methods to predict missing values.
    - Example: Using regression or k-NN to predict missing age.

Example Dataset:

ID	Name	Age	City	Marks
1	Ravi	20	Hyderabad	85
2	Sita	– (missing)	Vijayawada	78
3	Rohit	22	– (missing)	90
4	Anu	21	Hyderabad	– (missing)

After handling missing values:

ID	Name	Age	City	Marks
1	Ravi	20	Hyderabad	85
2	Sita	21	Vijayawada	78
3	Rohit	22	Hyderabad	90
4	Anu	21	Hyderabad	82

Explanation:

- Age of Sita is filled with average age: (20+22+21)/3 = 21.
- City of Rohit is filled with most frequent city (Hyderabad).
- Marks of Anu is filled with average marks: (85+78+90)/3 = 82.

Conclusion: *These techniques help us to handle missing values and improve the quality of data for better analysis.*

Q12. Explain normalization and standardization. Compare them with suitable examples and formulas.

Answer:

1. Normalization (Min-Max Normalization):

Normalization is the process of scaling numerical data into a fixed range, usually between 0 and 1. It keeps the relationships between values intact but changes the scale.

Formula: $X' = (X - Xmin) / (Xmax - Xmin)$

where X = original value, Xmin = minimum value in the dataset, Xmax = maximum value in the dataset.

Example: Let the marks of 3 students be 30, 60, 90. Here Xmin = 30, Xmax = 90.

For X = 60: $X' = (60 - 30) / (90 - 30) = 30/60 = 0.5$. So, 60 is normalized to 0.5.

2. Standardization (Z-score Standardization):

Standardization is the process of scaling data so that it has a mean of 0 and a standard deviation of 1. It is useful when data follows a normal distribution.

Formula: $Z = (X - \mu) / \sigma$

where X = original value, μ = mean of the dataset, σ = standard deviation of the dataset.

Example: Let the data be 20, 30, 40, 50, 60. Mean (μ) = 40, Standard Deviation (σ) ≈ 14.14.

For X = 50: $Z = (50 - 40) / 14.14 = 10 / 14.14 \approx 0.71$. So, 50 is standardized to 0.71.

3. Comparison:

Aspect	Normalization (Min-Max)	Standardization (Z-score)
Scale	Values between 0 and 1	Mean = 0 and Std. Dev = 1
Formula	(X − Xmin) / (Xmax − Xmin)	(X − μ) / σ
Use	When data is not normally distributed or scale is important	When data is normally distributed or algorithms assume normality
Outliers	Sensitive to outliers	Less sensitive to outliers
Example	Scaling age 10-50 to 0-1	Converting marks to Z-scores

> ***Conclusion:*** *Both techniques are used to scale data. Normalization keeps values in a fixed range, while Standardization centers data around zero with unit variance.*

Q13. What is feature engineering? Explain different feature engineering techniques with examples.

Answer:

Feature engineering is the process of creating new features from existing data to improve the performance of machine learning models. It helps in better representation of data, reduces overfitting and improves accuracy.

Different Feature Engineering Techniques with Examples:

1. **Creating New Features:** New features are created by combining or transforming existing features.
   - Example: If we have "Date of Birth", we can create an "Age" feature.
2. **Encoding Categorical Variables:** Categorical data is converted into numerical form.
   - Example: Gender (Male, Female) can be encoded as Male = 0, Female = 1.
3. **Handling Missing Values:** Missing values are filled using mean, median, mode or by creating a new feature.
   - Example: If "Age" is missing, fill it with average age or create an "Age_missing" feature (Yes/No).
4. **Feature Scaling (Normalization / Standardization):** Features are scaled to a similar range to improve model performance.
   - Example: Salary values (1000–100000) can be normalized to a range between 0 and 1.
5. **Feature Transformation:** Applying mathematical functions like log, square root to reduce skewness.
   - **Example: If "Income" is highly skewed, we can use log(Income).**
6. **Feature Selection:** Selecting important features and removing less useful or redundant features.
   - Example: Using correlation or variance threshold to select important features.
7. **Feature Interaction:** Creating interaction between two or more features.
   - Example: If we have "Price" and "Quantity", we can create a new feature "Total_Price" = Price × Quantity.

Conclusion: *Feature engineering helps in creating useful features from raw data, which improves the performance and accuracy of machine learning models.*

Q14. Explain feature scaling with suitable example.

Answer:

Feature scaling is the process of transforming numerical features so that they are on a similar scale. It is important in machine learning because features with large values can dominate the model and affect performance. The two commonly used techniques are Normalization (Min-Max Scaling) and Standardization (Z-score Scaling).

1. Normalization (Min-Max Scaling):

It scales the data to a fixed range, usually between 0 and 1.

Formula: $X' = (X - Xmin) / (Xmax - Xmin)$

Example: Suppose marks of a student in Maths are [30, 60, 90]. Here, Xmin = 30, Xmax = 90.

Original Marks (X)	Calculation	Normalized Value (X')
30	(30−30)/(90−30) = 0/60 = 0	0
60	(60−30)/(90−30) = 30/60 = 0.5	0.5
90	(90−30)/(90−30) = 60/60 = 1	1

Now the marks are scaled between 0 and 1.

2. Standardization (Z-score Scaling):

It transforms the data so that it has mean 0 and standard deviation 1.

Formula: $Z = (X - \mu) / \sigma$

Example: Suppose the age of students are [20, 30, 40]. Mean (μ) = (20+30+40)/3 = 30. Standard Deviation (σ) ≈ 8.16.

Original Age (X)	Calculation	Z-score (Z)
20	(20−30)/8.16 = −10/8.16	−1.22
30	(30−30)/8.16 = 0/8.16	0
40	(40−30)/8.16 = 10/8.16	1.22

Conclusion: *Feature scaling helps in faster convergence of algorithms and improves the accuracy and performance of machine learning models.*

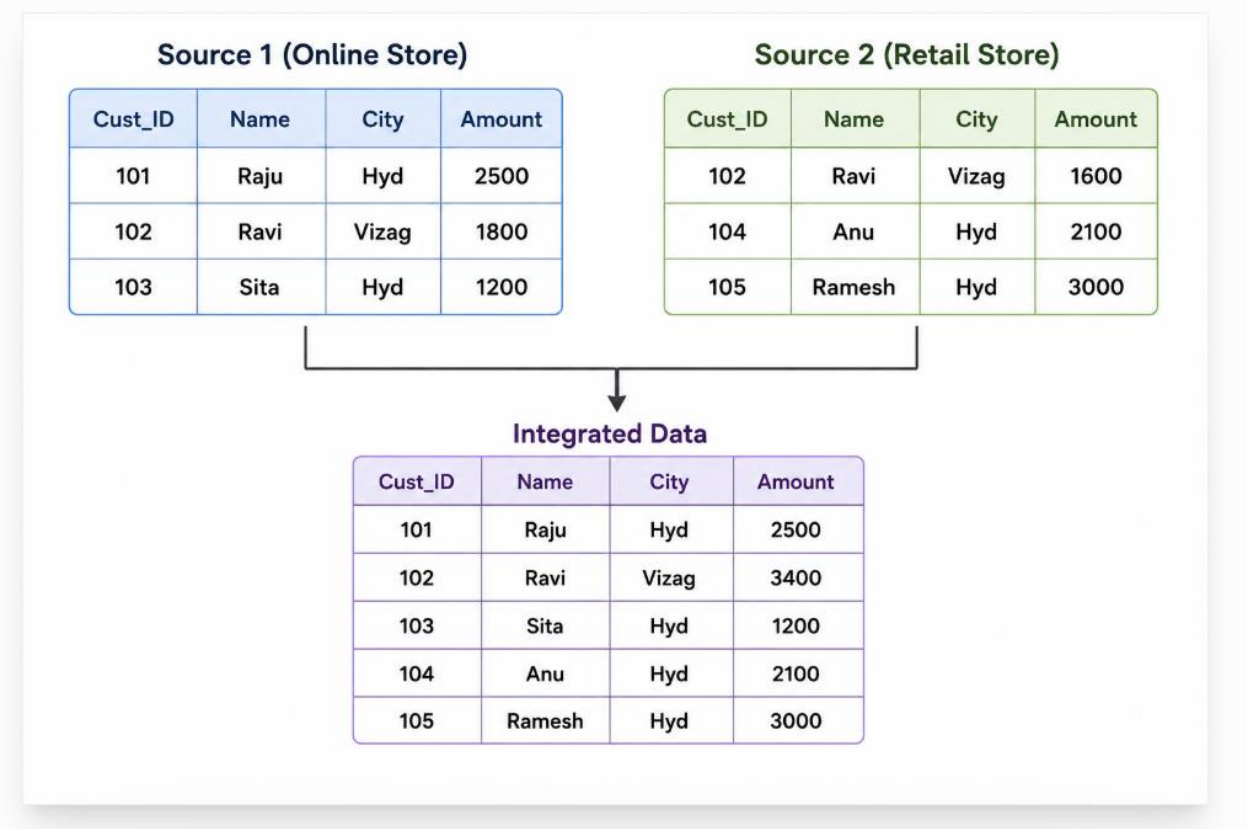
Q15. Explain data integration and data aggregation with suitable example.

Answer:

1. Data Integration:

Data integration is the process of combining data from different sources into a single, consistent and unified dataset. It helps to remove inconsistencies and provides a complete view of data.

Example: A company has customer information in two systems. We integrate both to get a single dataset.

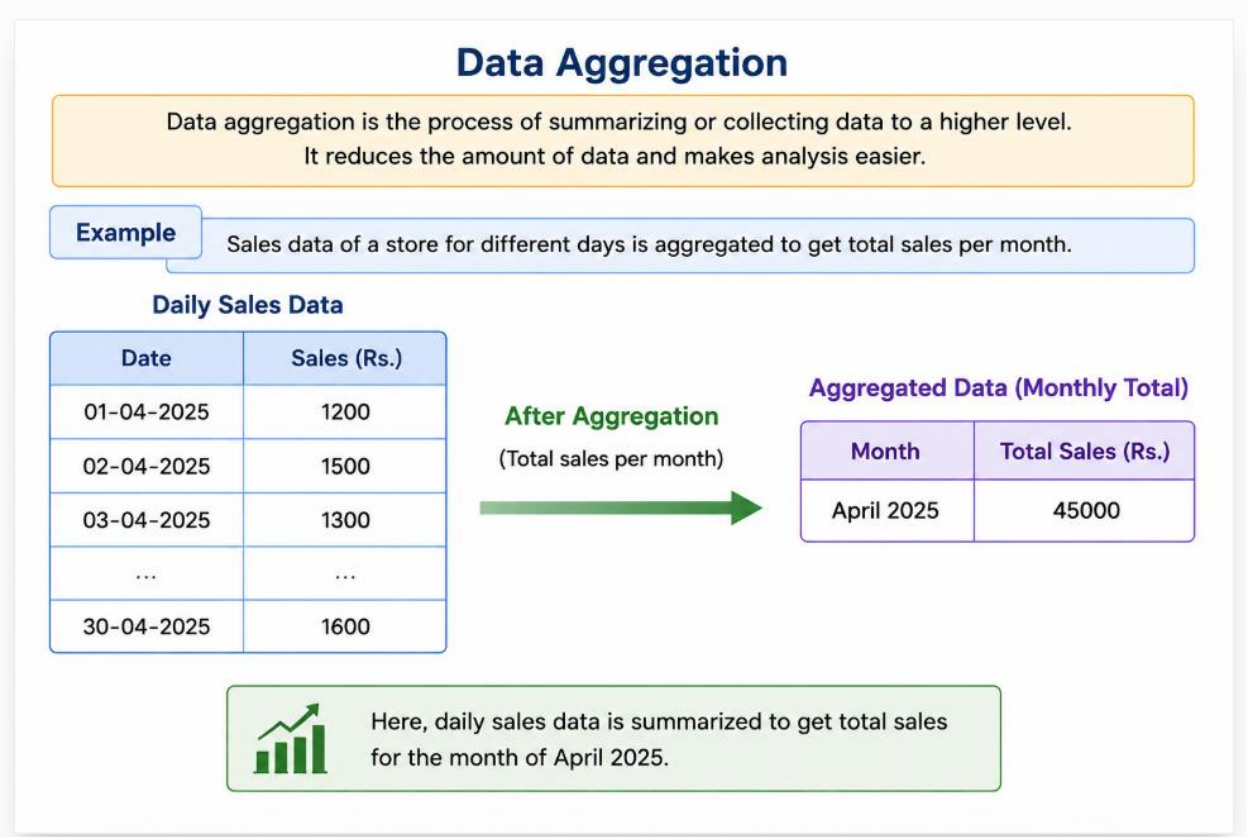


Here, data from both sources (Online Store and Retail Store) is combined and duplicate customer 102 is merged (1800 + 1600 = 3400).

2. Data Aggregation:

Data aggregation is the process of summarizing or collecting data to a higher level. It reduces the amount of data and makes analysis easier.

Example: Sales data of a store for different days is aggregated to get total sales per month.



Conclusion: *Data integration combines data from multiple sources, while data aggregation summarizes data to a higher level for easier analysis.*

Q16. For a given dataset, calculate the mean, median, mode, and standard deviation. Explain what each statistic represents.

Answer:

Dataset: 12, 15, 13, 18, 15, 20, 13, 17, 15, 12

1. Mean: Mean is the average of all values.

Formula: Mean ($\bar{x}$) = Σx / n

Calculation: Σx = 12+15+13+18+15+20+13+17+15+12 = 150, n = 10

Mean = 150 / 10 = 15

2. Median: Median is the middle value when data is arranged in order.

Ordered data: 12, 12, 13, 13, 15, 15, 15, 17, 18, 20

Median = (15 + 15) / 2 = 30 / 2 = 15

3. Mode: Mode is the value that appears most frequently.

In the data, 15 appears 3 times (most). Mode = 15

4. Standard Deviation: Shows how much the data values differ from the mean.

Formula (Population SD): $\sigma = \sqrt{[\Sigma(x - \bar{x})^2 / n]}$

x	12	15	13	18	15	20	13	17	15	12
$x - \bar{x}$	−3	0	−2	3	0	5	−2	2	0	−3
$(x - \bar{x})^2$	9	0	4	9	0	25	4	4	0	9

$\Sigma(x - \bar{x})^2 = 64$

$\sigma = \sqrt{(64/10)} = \sqrt{6.4} \approx 2.53$

What Each Statistic Represents:

- Mean: Gives the average or central value of the data.
- Median: Gives the middle value. It is not affected by extreme values.
- Mode: Gives the most common value in the data.
- Standard Deviation: Shows the spread of data. A smaller value means data is close to the mean. A larger value means data is more spread out.

Conclusion: *These statistics help in understanding and summarizing data.*

Q17. Explain how to identify and handle outliers in a dataset using suitable techniques such as filtering and discretization.

Answer:

Outliers are values that are very high or very low compared to other values in the dataset. They can affect the results of analysis. So, outliers must be identified and handled properly.

1. How to Identify Outliers?

Using Box Plot (IQR Method):

- Find Q1 (First quartile) and Q3 (Third quartile).
- Compute IQR = Q3 − Q1.
- Lower limit = Q1 − 1.5 × IQR.
- Upper limit = Q3 + 1.5 × IQR.
- Values outside these limits are outliers.

Using Z-score Method:

- Compute z = (x − mean) / standard deviation.
- If |z| > 3, the value is an outlier.

Example:

Dataset: 12, 13, 13, 14, 15, 15, 16, 18, 60

Q1 = 13, Q3 = 16 (from box plot). IQR = 16 − 13 = 3.

Lower limit = 13 − 1.5×3 = 8.5. Upper limit = 16 + 1.5×3 = 20.5.

Since 60 > 20.5, it is an outlier.

2. Techniques to Handle Outliers:

(A) Filtering (Removal or Capping):

In this method, outliers are removed or replaced with a limit value.

- Removal: Remove the outlier from the dataset.
- Capping (Winsorization): Replace the outlier with the nearest limit value (upper or lower limit).

Data Version	12	13	13	14	15	15	16	18	9th value
Original Data	12	13	13	14	15	15	16	18	60
After Removal	12	13	13	14	15	15	16	18	–
After Capping	12	13	13	14	15	15	16	18	20.5

(B) Discretization:

In this method, continuous data is converted into categories or intervals. Extreme values fall into the same interval and their effect is reduced.

Example: Age data

Age (Original)	Discretized Intervals	Explanation
0–20	0–20	Young
21–40	21–40	Adult
41–60	41–60	Middle age

Age (Original)	Discretized Intervals	Explanation
61–100 (outlier)	61–100	Old age (outlier effect reduced)

Conclusion: *Outliers can be identified using the IQR method or Z-score method. They can be handled using filtering (removal or capping) and discretization. These techniques improve the quality of data and help in accurate analysis and prediction.*